CRICKET PLAYER PERFORMANCE TRACKING APPLICATION : CRICKSTATS

USER STORY:

As a Salesforce Admin, I aim to develop a comprehensive player performance tracking application within Salesforce, designed to meticulously capture and analyze detailed, match-by-match performance data for cricket players. It will be built upon custom objects, such as Player and Match Performance, featuring precisely defined fields and relationships to guarantee optimal data integrity and accessibility. By strategically leveraging master-detail and lookup relationships, the application will empower the robust tracking of individual and team performances, facilitate insightful reporting and analytics, and streamline the comprehensive management of player statistics, ultimately enhancing strategic decision-making and fostering player development initiatives.

1. INTRODUCTION

1.1 PROJECT OVERVIEW :

CRICKSTATS is a Salesforce-based application meticulously crafted to track, manage, and analyze cricket players' performance across individual matches. Leveraging custom objects, intricate relationships, intuitive dashboards, and powerful automation tools, the system delivers structured and insightful performance analytics.

This empowers coaches, analysts, and administrators to effortlessly record player statistics, generate comprehensive reports, and confidently support strategic decision-making.

1.2 Purpose:

The primary objectives of this project are to:

- Digitize player performance records.
- Enhance the accuracy and accessibility of cricket statistics.
- Deliver real-time performance analytics.
- Facilitate informed player development and strategic team planning.
- Automate data management processes leveraging the Salesforce platform.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket team coach	Effectively track player performance across numerous matches	I struggle to obtain accurate statistics readily	Data is dispersed across disparate files and inconsistent formats	Frustrated and indecisive regarding optimal team selection
PS-2	A team manager	Efficiently maintain comprehensive player records	It consumes excessive time to update and verify data	The majority of records are still managed manually	Feeling stressed and operating inefficiently
PS-3	A sports analyst	Rigorously analyze player trends and generate insightful reports	I encounter significant difficulty in comparing performances	The absence of a centralized analytics system hinders this	Often feeling confused and ultimately dissatisfied with the analysis



2.2 Empathy Map Canvas

Target Users

- Team Managers
- Coaches
- Analysts
- Salesforce Admins

Empathy Analysis

Aspect	Description

- **Think & Feel:** Need accurate data, better insights
- **Hear:** Feedback from management
- **See:** Multiple data sources
- **Say & Do:** Collect statistics manually
- **Pain:** Data errors, delays
- **Gain:** Faster decisions, automation

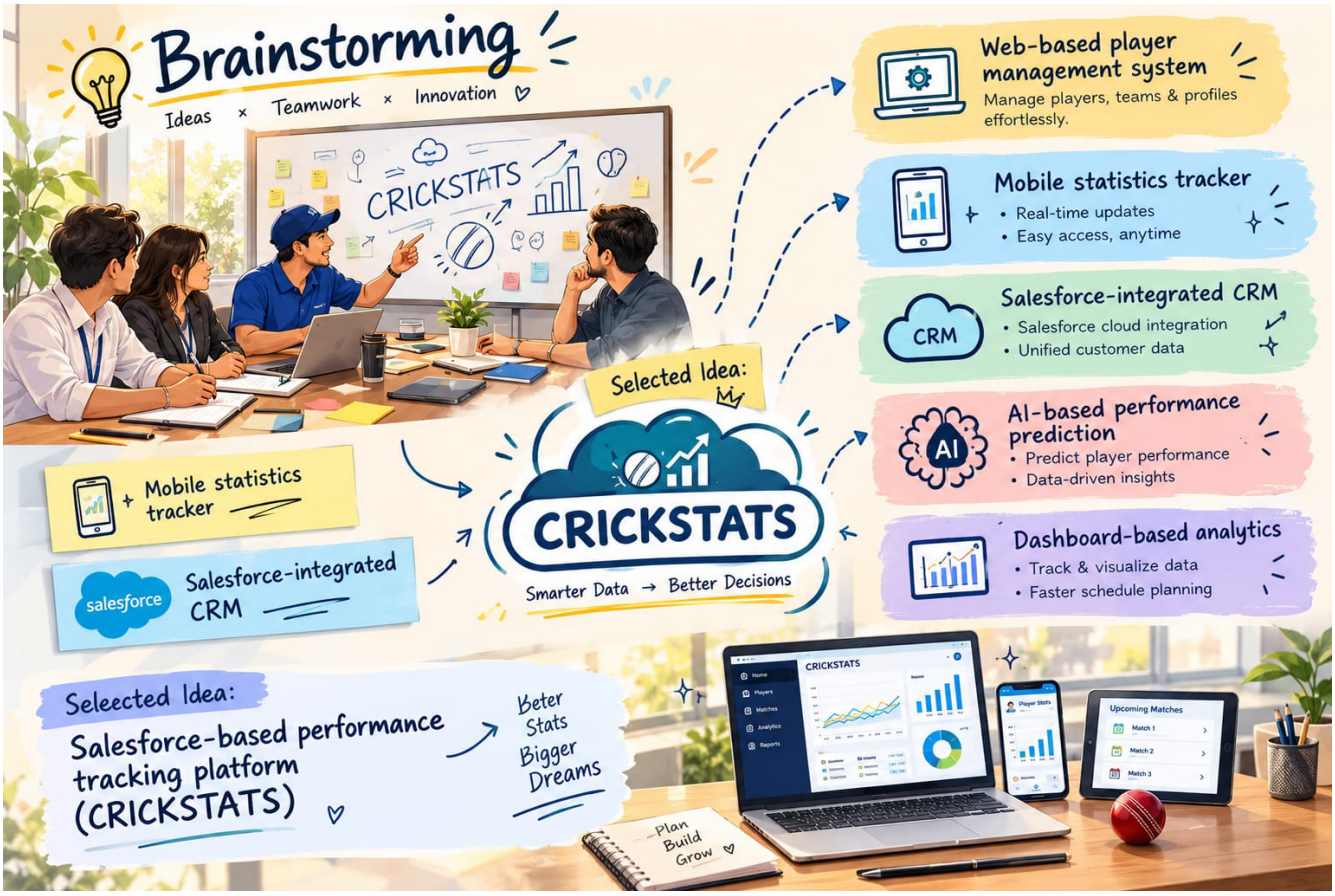


2.3 Brainstorming

Brainstormed Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Enter Player Data
- Record Match Performance
- Generate Reports
- Analyze Results
- Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements (NFRs) – CRICKSTATS

NFR-1: Usability

The system should be easy to understand and use by coaches, managers, analysts, and

admins without requiring technical training.

Requirements

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

Examples:

- A new coach should be able to:
- Log in
- Enter match data
- View player reports
- within a few minutes without special training.

Example: The Salesforce Lightning interface provides drag-and-drop dashboards and user-friendly forms.

NFR-2: Security

The system must protect sensitive player data from unauthorized access.

Requirements

- Role-based access control
- Secure authentication
- Encrypted data storage
- Regular backups

.Examples:

- Coaches can view player statistics
- Managers can edit records
- Analysts can generate reports
- Only admins can modify system settings

Example: Salesforce Profiles and Permission Sets restrict access to confidential information.

NFR-3: Reliability

The system should function correctly without failure and store data accurately.

Requirements

- No data loss
- Automatic backups
- Error handling
- System stability

Example:

If a manager enters match statistics and loses internet connection, the data should be saved automatically when reconnected.

Salesforce cloud backup ensures recovery in case of system failure.

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

Requirements

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal response time

Example:

When an analyst generates a performance report for 100+ players, the report should load within 2–3 seconds.

Optimized Salesforce queries and caching improve response speed.

NFR-5: Availability

The system should be accessible whenever users need it.

Requirements

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

Example:

Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends.

Salesforce provides high uptime with cloud hosting.

NFR-6: Scalability

The system should support increasing numbers of users, players, and records.

Requirements

- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

Example:

If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly.

Salesforce cloud infrastructure automatically scales storage and processing.

3.3 Data Flow Diagram

Data Flow

User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard

3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by

using secure cloud storage that safely stores all information and allows easy retrieval. Poor analysis of player performance, due to scattered and unorganized data, is addressed through interactive dashboards that present clear visual insights and reports. Additionally, delays in accessing performance information and generating reports are eliminated by providing real-time reporting features, enabling users to make faster and more informed decisions. Thus, CRICKSTATS successfully bridges the gap between existing problems and modern technological solutions.

4.2 Proposed Solution

Description

CRICKSTATS provides:

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

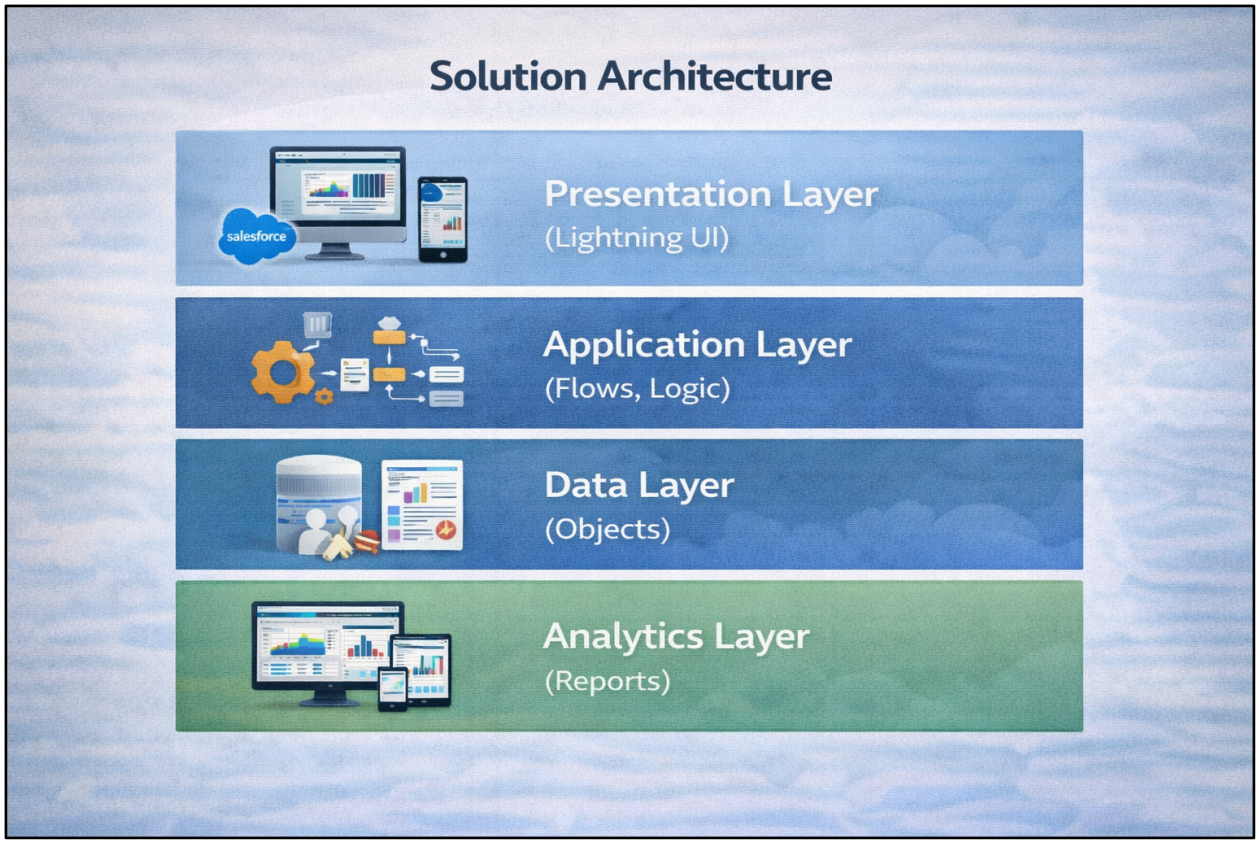
Key Features

- Centralized database
- Live statistics
- Secure access
- Scalable design
- Easy reporting

4.3 Solution Architecture

Architecture Layers

- Presentation Layer (Lightning UI)
- Application Layer (Flows, Logic)
- Data Layer (Objects)
- Analytics Layer (Reports)



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile-driven execution
- A seamless progression from object creation, through automation, to detailed reporting
- Development organized into distinct Sprints

Product Backlog, Sprint Schedule, and Estimation (4 Marks)
Utilize the template provided below for constructing the product backlog and sprint schedule.

Project Tracker, Velocity &Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-	20	6 Days	31 Oct	05 Nov 2022		

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
2			2022			
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. PROJECT DEVELOPMENT PHASE:

PROJECT FLOW

Milestone 1 : Creation of developer account
Milestone 2 : Object Creation
Milestone 3 : Tabs
Milestone 4 : The Lightning App
Milestone 5 : Fields & Relationships
Milestone 6 : Roll-up Summary
Milestone 7 : Flows

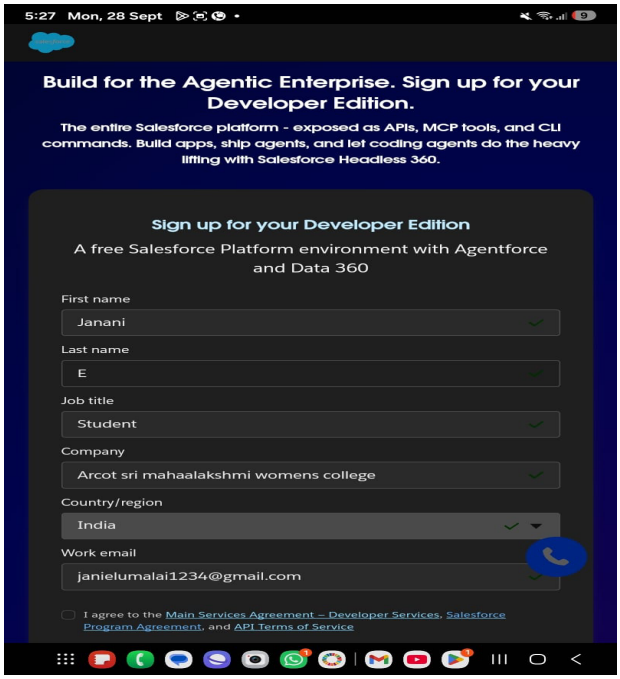
IMPLEMENTATION

Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :



1) First name & Last name
2) Email
3) Role : Developer
4) Company : College or Company Name
5) County : India
6) Postal Code : pin code
7) Username : should be a combination of your name and company

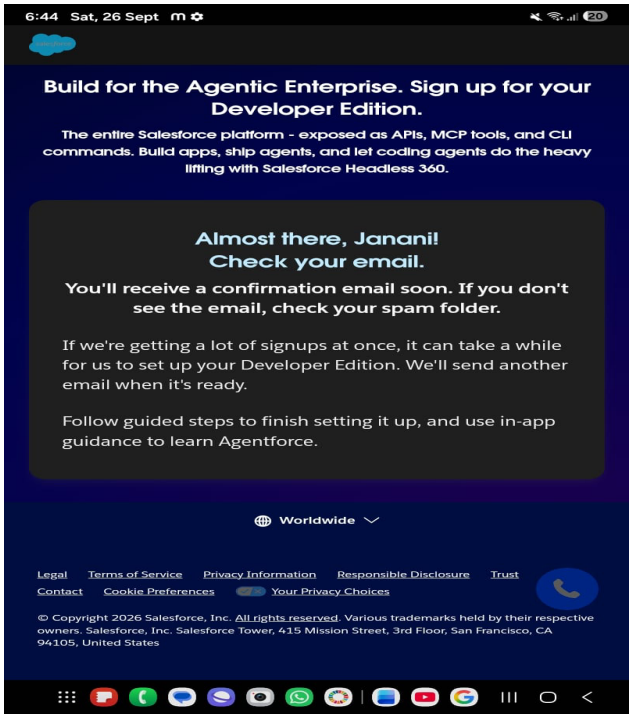
This need not be an actual email id, you can give anything in that format : username@organization.com

Click on sign me up after filling these.

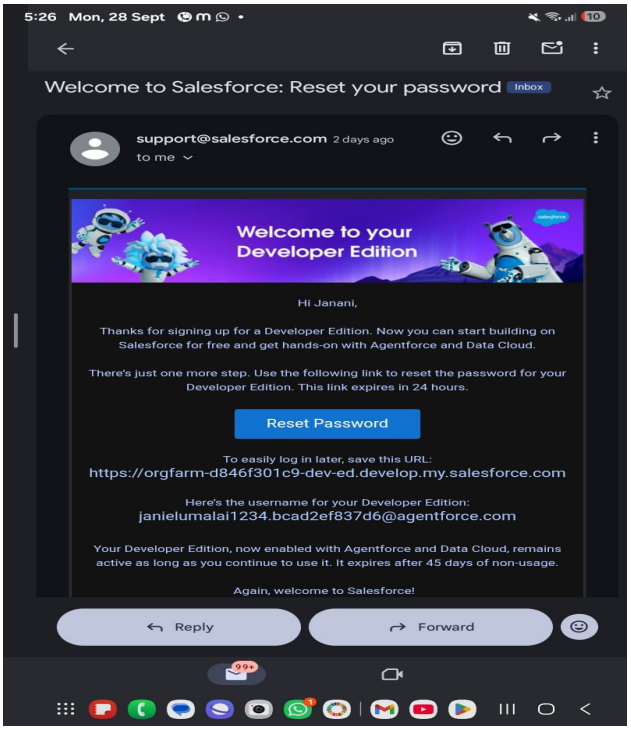
Activity 2: Account Activation :

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

2. Click on Verify Account.



3. Give a password and answer a security question and click on change password.



4. Then you will redirect to your salesforce setup page.

Milestone 2 - Object Creation

Activity 1 : Player Object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. On the Object Manager page, click Create | Custom Object.

From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label** : Player
- **Plural Label** : Players
- Enter Record Name Label and Format
  Record Name : Player Name
  Data Type **:** Text
- **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status :** check the box for Allow Search.
- Save

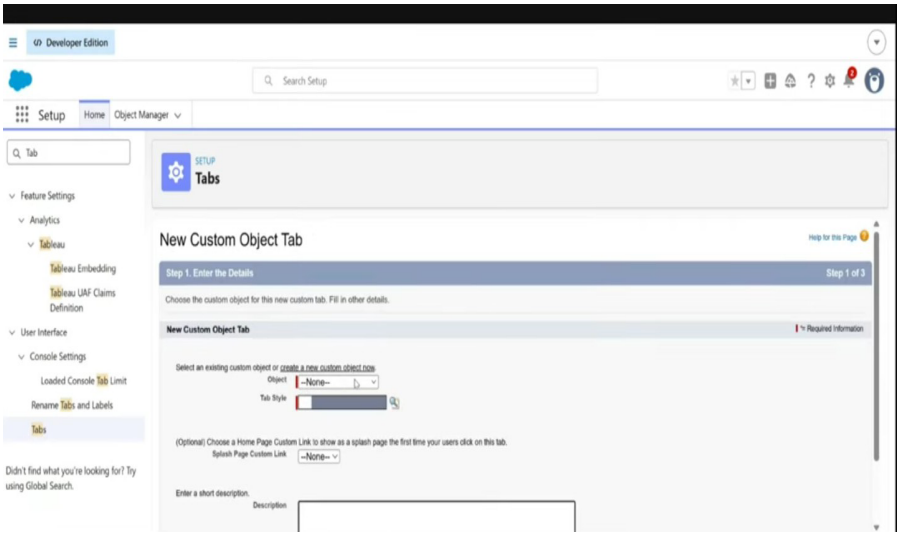
Activity 2 : Match Performance Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
   - **Label** : Match Performance
   - **Plural Label** : Match Performances
   - Enter Record Name Label and Format
     Record Name : Customer Account Name
     Data Type **:** Text
   - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
   - **Search Status :** check the box for Allow Search.
   - Save.

Milestone 3 - Tabs

Activity 1 : Custom tabs creation

1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)
2. Select Object(Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Match Performances.



Milestone 4 - The Lightning App

Activity 1 : Create a lightning app

Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning app.

1. Fill the app name in app details and branding as follow
   App Name : CRICKSTATS
   Developer Name : Auto Populates
   Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default.
   Then click Next.

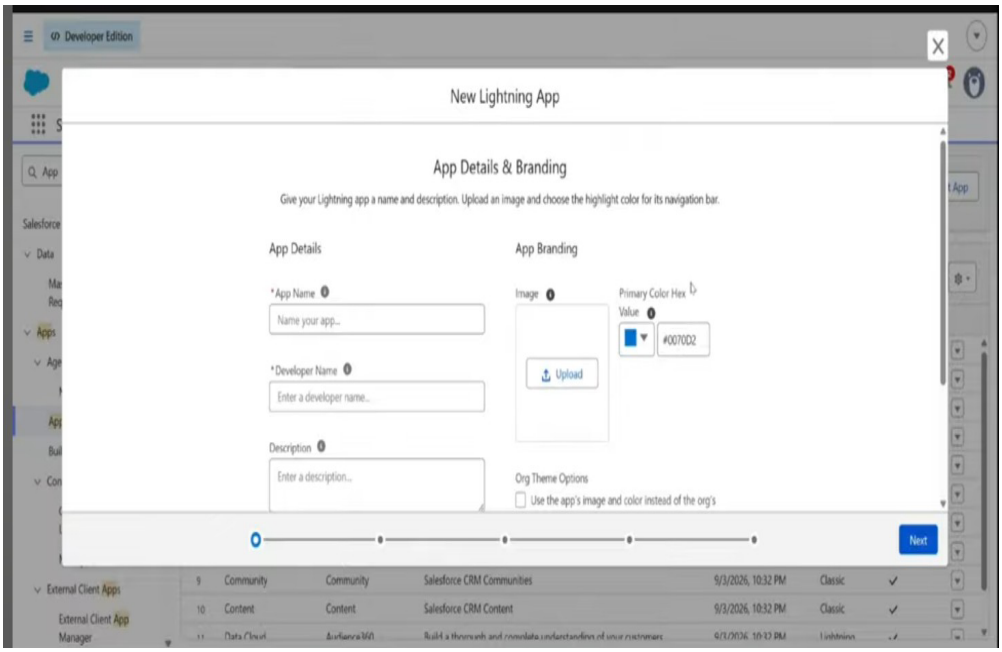
2. In App options
   Navigation style : Standard navigation

Setup experience : setup
Supporters form factors : Desktop and phone
Then click Next

3. In Utility items
   Utility Bar alignment : Default
   Then click Next.

4. Navigation items
   Select the created Custom Objects and required standard objects
   - Players
     Match Performances
     Reports
     Dashboards

5. To Add User Profiles : System Administrator



6. Click Save & Finish.

Milestone 5 : Fields & Relationships

Activity 1 : Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input. Click on next.

7. Data Type : **Text**
   Field Label: Player Name
   Field Name : Auto Populates
   Length : 50
   Click on Next → Next → Save and new.

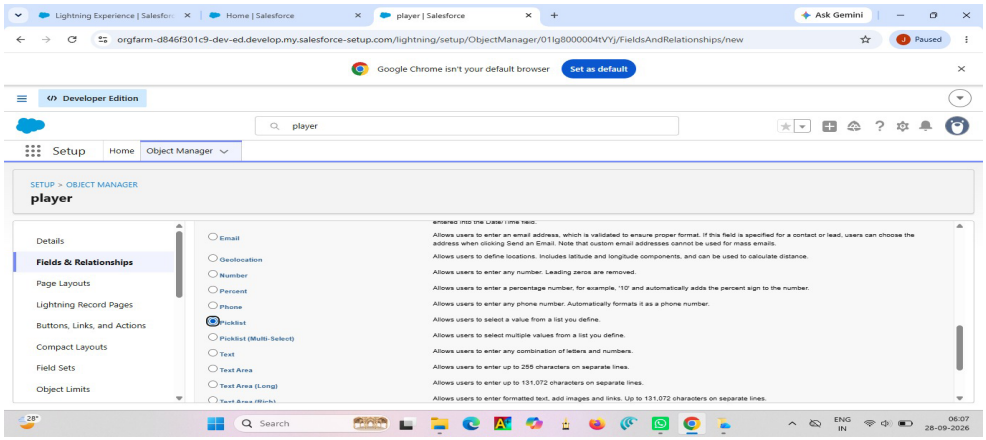
8. Repeat the Same steps for remaining fields

   - Data Type : **Text**
     Field Label: Player Number
     Field Name : Auto Populates
     Length : 3
     Click on Next → Next → Save and new.

   - Data Type : **Rich Text Area**
     Field Label : Profile Picture
     Field Name : Auto Populates
     Click on Next → Next → Save and new.

   - Data Type : **Date**
     Field Label : Date of Birth
     Field Name : Auto Populates
     Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Role
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Batsman
  Bowler
  Batsman & Wicket Keeper
  All Rounder



- Data Type : **Picklist**
  Field Label : Batting Style
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Right- Handed
  Left - Handed
  Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Bowling Style
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Right-Arm Fast
  Left-Arm Fast
  Right-Arm Medium
  Left-Arm Medium
  Right-Arm Spin
  Left-Arm Spin
  Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Nationality
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Australia
  Bangladesh
  England
  India
  New Zealand
  Pakistan
  South Africa
  Sri Lanka
  West Indies
  Afghanistan
  Zimbabwe
  Ireland
  Scotland
  Netherlands
  United Arab Emirates
  Nepal
  United States of America
  Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Team
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line
  Chennai Super Kings (CSK)
  Delhi Capitals (DC)
  Kolkata Knight Riders (KKR)
  Mumbai Indians (MI)
  Punjab Kings (PBKS)
  Rajasthan Royals (RR)
  Royal Challengers Bangalore (RCB)
  Sunrisers Hyderabad (SRH)
  Gujarat Titans (GT)
  Lucknow SuperGiants (LSG)
- Data Type : **Formula**
  Field Label : AVG Score
  Field Name : Auto Populates
  Formula Return : **Number**
  Formula : IF(
  Matches_Played__c > 0,
  Total_Runs__c / Matches_Played__c ,

0
)

Click on Next → Next → Save & New.

- Data Type : **Lookup Relationship**
  Related To : Match Performance
  Field Label : Match Performance
  Field Name : Auto Populates
  Click on Next → Next → Save.

Activity 2 : Create Custom fields for Match Performance object.

1. Data Type : **Text**
   Field Label: Match Performance Name
   Field Name : Auto Populates
   Length : 80
   Click on Next → Next → Save and new.

2. Data Type : **Checkbox**
   Field Label: Player of the Match
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

3. Data Type : **Picklist**
   Field Label: Match Result
   Field Name : Auto Populates
   Value : Enter value, with each value separated by a new line
   WON
   LOST
   DRAW
   Click on Next → Next → Save and new.

4. Data Type : **Number**
   Field Label: Runs Scored
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

5. Data Type : **Number**
   Field Label: Balls Faced
   Field Name : Auto Populates

Click on Next → Next → Save and new.

6. Data Type : **Number**
   Field Label: Fours
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

7. Data Type : **Number**
   Field Label: Sixes
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

8. Data Type : **Number**
   Field Label: Overs Bowled
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

9. Data Type : **Number**
   Field Label: Wickets Taken
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

10. Data Type : **Number**
    Field Label: Runs Conceded
    Field Name : Auto Populates
    Click on Next → Next → Save and new.

11. Data Type : **Number**
    Field Label: Maiden Overs
    Field Name : Auto Populates
    Click on Next → Next → Save and new.

12. Data Type : **Number**
    Field Label: Stumpings
    Field Name : Auto Populates
    Click on Next → Next → Save and new

13. Data Type : **Number**
    Field Label: Catches
    Field Name : Auto Populates
    Click on Next → Next → Save and new.

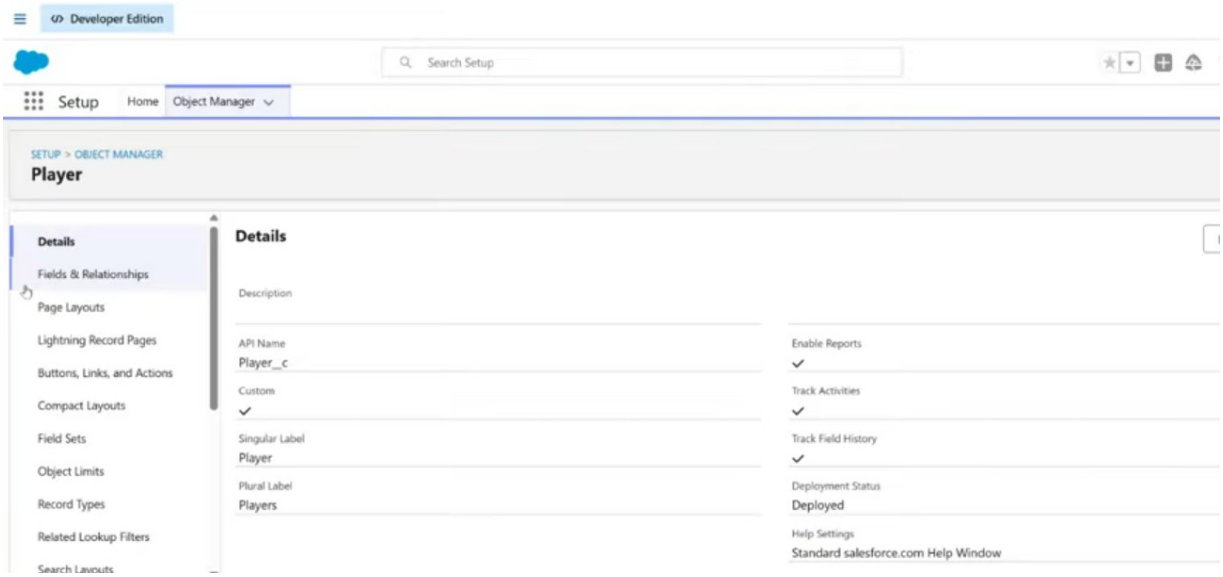
14. Data Type : **Master - Detail Relationship**
    Related To : Player
    Field Label : Player
    Field Name : Auto Populates
    Click on Next → Next → Save.

Milestone 6 : Roll-up Summary

Activity 1 : Create Roll-up Summary for Player object.

Note :- Roll-up Summary can only be created on parent object in Master- Detail Relationship.

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**



4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,
6. Choose a data type as **Roll-up Summary**. Click Next.

Field Label: Matches Played
Field Name : Auto Populates
Summarized Object : Match Performances
Select Roll-up Type : COUNT

Filter Criteria : All records should be included in the calculation.
Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Number of 50’s
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Number of 100’s
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total 4’s
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Fours
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total 6’s
  Field Name : Auto Populates

Summarized Object : Match Performances
Select Roll-up Type : SUM
Field To Aggregate : Sixes
Filter Criteria : All records should be included in the calculation.
Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Runs
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Runs Scored
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Overs
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Overs Bowled
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Wickets
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Wickets Taken
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Stumpings
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Stumpings
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Catches
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Catches
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

Milestone 7 : Flows

Activity 1 : Screen Flow

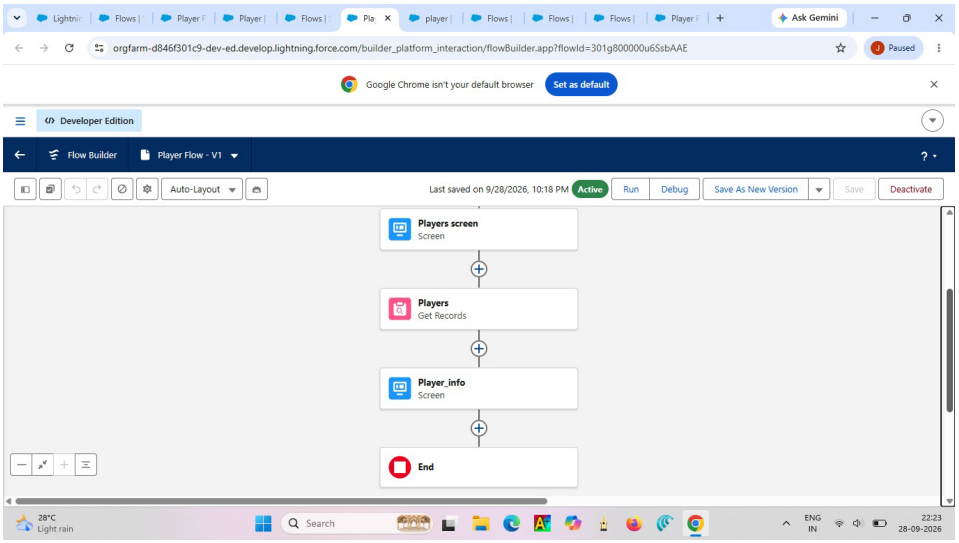
1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.Select Screen Flow and click Create.

2. Click (+) Icon on the flow builder and Select **Screen Component.**
   Label : Players screen
   API : Players_screen
   Select the **Picklist** component and place it on the screen.
   Label : Team
   API : Team
   From Component Pane on the left side select **Component Type** : Picklist
   Data Type : Text
   Choice : Click New Choice Resources
   Resource Type : Picklist Choice Set
   API Name : Team
   Object : Player
   Data Type : Picklist
   Field : Team
   Click Done.

3. Click (+) Icon on the flow builder and Select **Get Records Component.**
   Label : Players
   API Name : Players
   Object : Player
   Filter player records :

Field : Team
Operator : Equals
Value : Click New Resource
Resource Type : Variable
API Name : Dynamic
Data Type : Text
Default Value : Screen component : Team (Picklist)

4. Click (+) Icon on the flow builder and Select **Screen Component.**
   Label : Player_info
   API Name : Player_info
   From Component Pane on the left side select **Component Type :** Data table
   API Name : Player_info_table
   Label : Data Table
   Configure Data Source : Source Collection : Players from players (Player__c)



In Configure Columns : Add all the required Columns
Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style | Matches Played | Number of 50's | Number of 100's
Click Done.

Click Save.
Flow Label : Player Flow

Flow API Name : Player_Flow

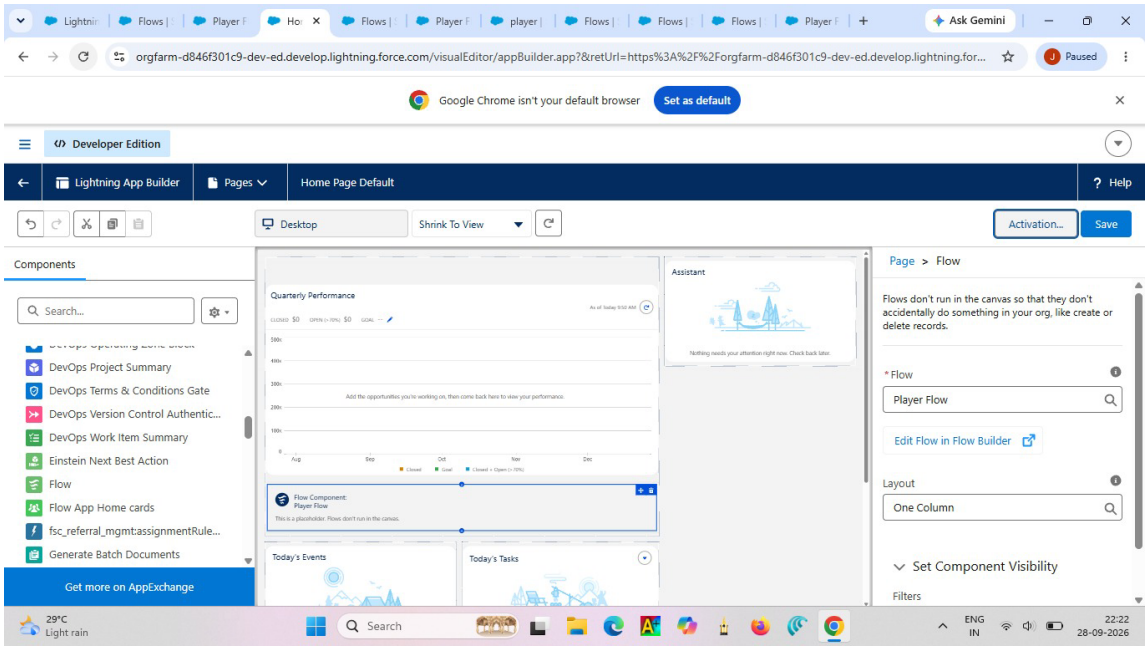
Activity 2 : Flow on Home Page

1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.

2. Remove all other components and Add **FLOW** component from the standard components on the left side.

3. **Flow** : Players Flow

4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to Apps.**



6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.

7. FUNCTIONAL & PERFORMANCE TESTING

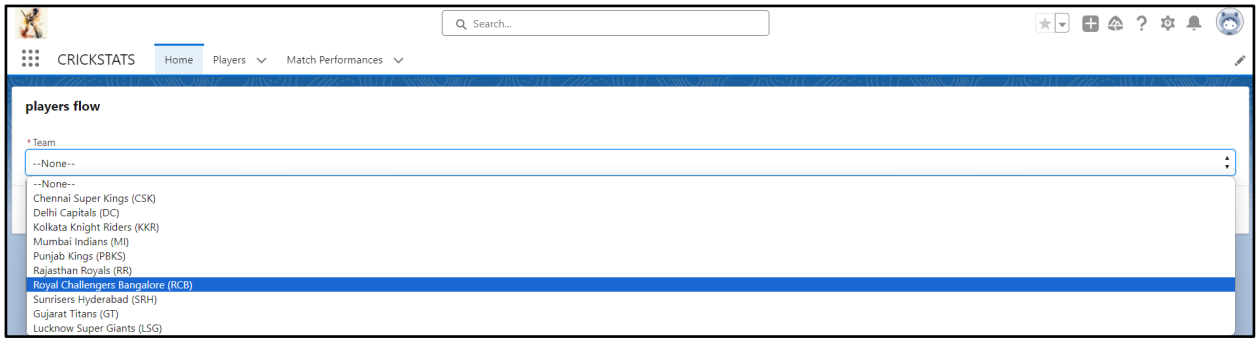
During the testing phase, screenshot were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for

Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

NOTE : Enter some records in the Player Object and Match Performance Object to get the results.

1. Select one Team from the Picklist and Click **Next**.



8.ADVANTAGES &DISADVANTAGES

Advantages

- Unified Centralized System
- Dynamic Real-time Analytics
- Robust Secure Platform
- Highly Scalable Architecture

- Streamlined Automation and Reduced Manual Effort

Disadvantages

- Requires Salesforce License
- Dependence on Internet Connectivity
- Limited Offline Access

9. CONCLUSION:

CRICKSTATS has successfully implemented a sophisticated cloud-based cricket performance tracking system, strategically leveraging Salesforce. This innovative system automates data management, enhances reporting accuracy, and effectively facilitates informed decision-making. The project distinctly demonstrates the practical application of SDLC principles, Agile methodology, and CRM-based development.

10. FUTURE SCOPE

- Seamless Mobile Application Integration
- Advanced AI-driven Performance Prediction
- Comprehensive Video Analysis Integration
- Dynamic Player Ranking System
- Interactive Public Statistics Portal
- Expanded Multi-League Support

11. APPENDIX

- Detailed Salesforce Objects and Fields
- Manual Match Statistics Documentation
- System Flow Configurations